

Exercise Session 11: Magnetic Fields in Matter 2 and Magnetostatics Recap

Question 1 (From Section 6.1.2):

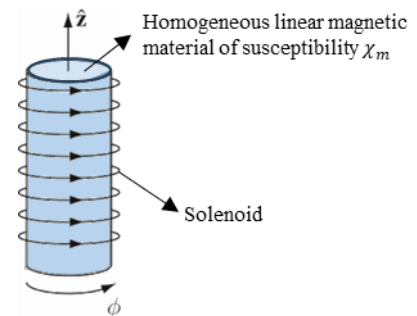
Answer the following:

- a) Why can a magnetic dipole sometimes be pictured as if it had north and south poles (Gilbert model), even though isolated magnetic poles are not physically used in the real model?
- b) Why is a current loop (Ampèrian model) a better physical model for a magnetic dipole than a pair of separated magnetic charges?
- c) In what sense is a bar magnet similar to an electric dipole, and in what sense is the analogy incomplete?

Question 2 (From Section 6.4.1):

An infinitely long solenoid with n turns per unit length carries a steady current I . The interior of the solenoid is completely filled with a homogeneous linear magnetic material of susceptibility χ_m .

- (a) Define magnetic susceptibility, and write the constitutive relations connecting $\mathbf{M}$, $\mathbf{H}$, and $\mathbf{B}$ for a linear magnetic medium.
- (b) Explain why, for this system, $\mathbf{H}$ can be determined directly from the free current, whereas $\mathbf{B}$ is not usually calculated first.



- (c) State how the magnetic field inside the solenoid changes when the material is:

- I. paramagnetic,
- II. diamagnetic.

Give a physical explanation in each case.

- (d) The bound surface current density is given by

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}.$$

State the direction of $\mathbf{K}_b$ relative to the solenoid current for:

- I. $\chi_m > 0$,
- II. $\chi_m < 0$.

(e) In a homogeneous linear medium,

$$\mathbf{J}_b = \nabla \times \mathbf{M} = \chi_m \mathbf{J}_f.$$

Explain the physical meaning of this relation. Under what condition will the bound current exist only on the surface of the material?

Question 3 (Bound currents, magnetic susceptibility and permeability):

A very long coaxial cable consists of two thin cylindrical conducting shells of radii a and b ($a < b$). The region $a < s < b$ between the conductors is filled with a homogeneous linear magnetic insulator of susceptibility χ_m . A steady current I flows along the inner conductor in the $+z$ -direction and returns along the outer conductor in the $-z$ -direction. In both conductors, the current is uniformly distributed over the surface.

- (a) Find the magnetic field intensity $\mathbf{H}$ and magnetic flux density $\mathbf{B}$ in the region $a < s < b$.
- (b) Determine the magnetization $\mathbf{M}$ in the magnetic medium.
- (c) Calculate the bound volume current density $\mathbf{J}_b$ and the bound surface current densities at $s = a$ and $s = b$.
- (d) Show that the free and bound currents together produce the same magnetic field found in part (a).

Question 4 (Bound currents, magnetic susceptibility and permeability):

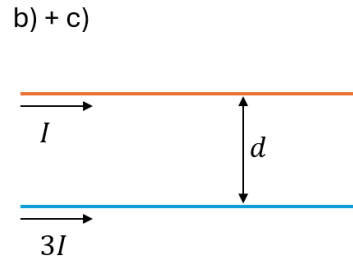
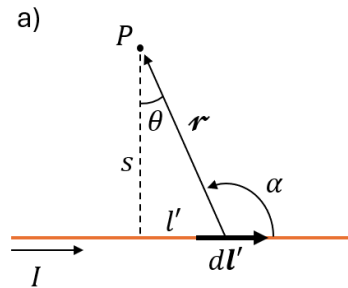
A long cylindrical wire of radius a is made of a homogeneous linear magnetic material of susceptibility χ_m . It carries a steady current I , uniformly distributed over its cross-section, in the $+z$ -direction.

- (a) Find the magnetic field intensity $\mathbf{H}$ and magnetic flux density $\mathbf{B}$ for the regions $s < a$ and $s > a$, where s is the distance from the axis.
- (b) Determine the magnetization $\mathbf{M}$ inside the wire.
- (c) Find the bound volume current density $\mathbf{J}_b$ and the bound surface current density $\mathbf{K}_b$.
- (d) Determine the net bound current flowing along the wire.

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Question 5 (Biot-Savart law):

- a) Find the magnetic field a distance s from an infinite straight wire carrying a steady current I (see figure) using the Biot-Savart law.
- b) Find the magnetic field due to the above wire at a second infinite straight wire placed a distance d away from and parallel to the wire in a), see figure.
- c) Find the Lorentz force per unit length between the two infinite straight wires in b), if the second wire carries a steady current $3I$.



Hint: Exploit the geometry and the definitions provided in the figure. A bit of trigonometry comes in handy.